

Machine Learning for Solving Charging Infrastructure Planning: A Comprehensive Review

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Abstract—The ever-growing energy demand accompanied with environmental pollution has initiated a paradigm shift towards Electric Vehicles (EVs) from conventional vehicles. Public acceptance of EVs call for availability of charging infrastructure. Charging infrastructure planning is an intricate process involving various activities such as charging station placement, charging demand prediction, charging scheduling etc and interaction of power distribution as well as road network. In recent years, the advent of machine learning has made data driven approaches popular for solving charging infrastructure planning problem. Consequently, researchers have started using machine learning techniques for solving problems associated with charging infrastructure planning such as charging station placement, charging demand prediction, charging scheduling etc. This work aims to provide a comprehensive review of machine learning applications for solving charging infrastructure planning.

Keywords—electric vehicle, charger, machine learning, review

I. INTRODUCTION

The consumption of energy is increasing globally at an alarming rate and it is observed that the transport sector is one of the largest consumers of energy [1]. It has been found that in US, approximately 28% of net energy consumption in 2019 was for transporting people and goods from one place to another [2]. Further, it has been found that the transport sector is one of the major agents of air pollution [3]-[5]. The paradigm shift from Internal Combustion Engine (ICE) driven vehicles to EVs is a viable way to tackle the serious concerns of energy crisis and air pollution. Large scale adoption of EVs require availability of charging infrastructure. Planning of charging infrastructure involves the interaction of both road and power distribution network. Placement of charging stations at the weak points of the power distribution network and uncoordinated charging may be detrimental and result in voltage instability, increased power losses, harmonic distortions, degraded reliability indices [6]-[12]. Further, the charging infrastructure planning must also take into account the EV drivers convenience such as accessibility of the charging stations, waiting time in the charging stations etc [13]. Also, smart coordinated charging is preferred over uncoordinated charging to tackle the detrimental impact of EV charging on grid [14].

The charging infrastructure planning is a multi-faceted problem involving a number of decision variables, objective functions, and constraints. Researchers have used heuristics

[15], [16], meta-heuristics [17], machine learning [18], and game theory [19], [20] for solving this problem. In recent years, the advent of machine learning has made data driven approaches popular for solving charging infrastructure planning problem. Consequently, researchers have started using machine learning techniques for solving problems associated with charging infrastructure planning such as charging station placement, charging demand prediction, charging scheduling etc. This work aims to provide a comprehensive review of machine learning applications for solving charging infrastructure planning. Table I lists some of the prominent research works that meticulously reviews different aspects of charging infrastructure planning. From Table I, it is observed that researchers have reviewed various aspects of e mobility such as charging station placement, drivers for EV adoption, policies for promoting EVs, charging technologies, charge scheduling etc. However, it is observed that there is lack of comprehensive and systematic reviews reporting the applications of machine learning for solving the charging infrastructure planning problem. Hence, this work aims to provide a comprehensive review of machine learning applications for solving charging infrastructure planning.

II. CHARGING INFRASTRUCTURE PLANNING

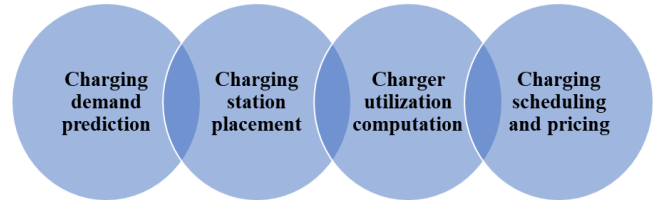


Fig. 1. Different activities of charging infrastructure planning

Charging infrastructure planning is a prerequisite for large scale adoption of EVs. The different activities associated with charging infrastructure planning are as shown in Fig. 1. Charging demand prediction involves prediction of the demand of charging service during different times of the day and at different location. Charging station placement is a typical planning problem seeking an optimal allocation and sizing of charging stations considering economic factors, operating parameters of distribution network and EV driver's convenience. Charger utilization computation involves computing how much a charger is utilized or how many charging events a charger has served. Charging scheduling involves managing the charging activities based on charging

demand and load profile keeping in mind that the power grid must not be overloaded.

TABLE I. REVIEW WORKS ON CHARGING INFRASTRUCTURE PLANNING

Ref	Year	Diligence
[21]	2018	Review of consumer preferences of and interactions with EV charging infrastructure.
[22]	2019	Review of spatial localization methodologies for the electric vehicle charging infrastructure.
[23]	2018	Review of economics of charging infrastructure planning
[24]	2018	Review of fast charging infrastructure for EVs
[25]	2020	Review of EV charging standards and grid impacts of EV charging
[26]	2018	Review of policies, methodologies, and challenges for charging infrastructure deployment in China
[27]	2017	Review of factors affecting adoption of EVs
[28]	2016	Review of recent trends in optimization techniques for plug-in hybrid, and electric vehicle charging infrastructures
[29]	2018	Suggestion on tax policy for promoting the PPP projects of charging infrastructure in China
[30]	2019	Review of world wide policy measures to promote e mobility
[31]	2018	Review of the electric vehicle charging techniques, standards, progression and evolution of EV technologies in Germany
[32]	2018	Review of global EV diffusion model
[33]	2020	Review on approaches for EV charging demand management
[34]	2017	Review of EV policies in China
[35]	2018	Review of EV DC charging stations using photovoltaic sources.
[36]	2017	Review of EV industrialization in China
[37]	2019	Review of financial incentives for EV adoption
[38]	2018	Review of metaheuristics for solving charging scheduling problem
[39]	2019	Review on power interaction mode, scheduling methodology and mathematical foundation for EV integration with power grid
[40]	2020	Review of current scenario of EV charging service planning and operation considering transport and power network
[41]	2020	Review of strategic charging–discharging control of grid-connected electric vehicles
[42]	2018	Review of EVs charging from the perspective of energy optimization, optimization approaches, and charging techniques
[43]	2019	Review of dynamic pricing for EVs in charging stations
[44]	2019	Review of power quality improvement in smart grids by EVs
[45]	2019	Review of planning of grid connected charging stations
[46]	2020	Review on charging behavior of data, model, and control in EV charging stations
[47]	2018	Review of static and dynamic wireless electric vehicle charging system
[48]	2018	Review of solar EV charging station
[49]	2018	Review of wireless charging for smart cities
[50]	2018	Review of Level 2 charging system for EVs

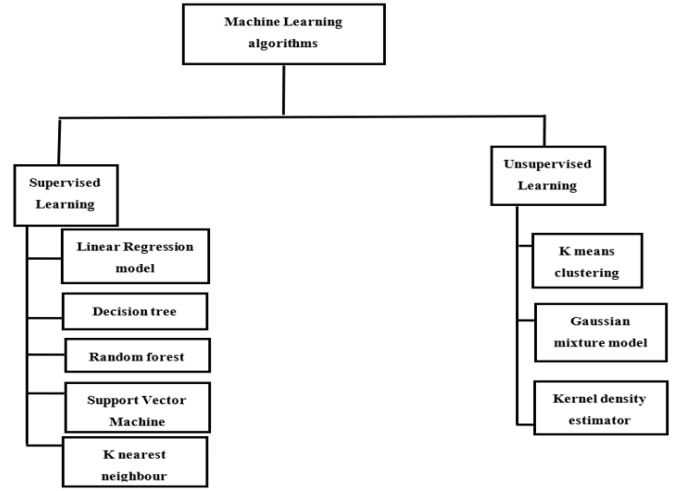


Fig. 2. Classification of machine learning algorithms

III. MACHINE LEARNING TECHNIQUES

In machine learning the computer learns from previous experience without any explicit programming [51]. In this context, experience refers to the dataset that the algorithm uses to train themselves [52]. With time and learning experience, the models can predict accurately the trends thereby providing predictive analysis [51]. Typically, machine learning algorithms are categorized into supervised and unsupervised learning [51], [53], [54]. Further, depending on the type of variable machine learning algorithms can be divided into regression problem and classification problem [51]. If the response variable is continuous then it is called a regression problem [51]. On the other hand, if the response variable is categorical then it is called a classification problem [51]. The classification of the machine learning algorithms are as shown in Fig.2. A detailed description of these classifications is provided in the subsequent sub sections.

A. Supervised Learning

As the name indicates, in supervised learning machine learning models are trained by labelled datasets [51], [55], [56]. The dataset contains input variable as well as the target variable. The learning of the model is iterative in nature by mapping between input and target output assisted by optimization [51]. As shown in Fig.2, supervised learning can be divided into 5 types. In linear regression model, there is a linear relationship between input variable and target variable [51]. As indicated by the name linear regression can be used for regression problems and for linearly separable datasets [54]. Decision tree can be used for both regression as well as classification problems [54]. Decision trees separate complex decisions into mix of simpler decisions using split points [54], [57], [58]. In random forest several decision trees are aggregated for the purpose of prediction [59], [60]. Support Vector Machine (SVM) is mainly used for classification problem but can also be utilized for regression problems [61], [62]. SVM separates the classes with best hyperplane that can maximize the marginal difference between the classes [18], [62]. The training time for SVM is long and therefore it is not suitable for large datasets [61], [62]. K nearest neighbor (KNN)

can be used for both regression and classification problems [18], [63], [64]. However, it is mostly used for classification problems [18]. KNN does not require a dedicated training phase and a lazy learning phase is associated with it [18], [63], [64].

B. Unsupervised Learning

In case of unsupervised learning, the training dataset comprises of input variable only [18], [65], [66]. The key goal of this model is to find patterns within the dataset by clustering [18], [65], [66]. The subdivisions of unsupervised learning are as illustrated in Fig. 2. In k means of clustering, individual data points form k clusters where each and every point is assigned to k center points at the starting in a random fashion [18], [67] and later data points are assigned to nearest center based on new data point computation. Gaussian mixture model (GMM) is a probabilistic learning model that has the capacity to represent subpopulations with normal distribution by considering multiple normal distributions of the dataset in use [18]. Kernel Density Estimator (KDE) is used in case of non-parametric probability density function [18].

IV. MACHINE LEARNING FOR CHARGING INFRASTRUCTURE PLANNING

The applications of machine learning techniques for solving different charging infrastructure planning problem shown in Fig. 1 is illustrated in this section.

A. Machine Learning for Charging Station Placement

Charging station problem involves finding the location and size of chargers to be placed at that location. In [68], the authors provided an optimal wireless charging station placement scheme for electric trams by applying an algorithm considering hybridization of Genetic Algorithm (GA) with Reinforcement Learning (RL). The integration of GA with reinforcement learning improved the performance of GA thereby preventing it from getting stuck in local optima. The superior performance of the hybrid GA RL algorithm with the standalone algorithm is as illustrated in Table II.

TABLE II. PERFORMANCE COMPARISON OF HYBRID GA AND RL WITH STANDALONE ALGORITHMS [68]

Parameter	GA	RL	GA+RL
Investment cost (\$)	106.230	104.561	103.891
Battery capacity (kWh)	15.06	13.14	12.60

In [69], the authors provided a novel placement scheme for placing new charging stations by considering maximization of utilization rate of chargers as the objective function. The problem was solved by hierarchical clustering [70]. In [71], the authors categorized the charging stations as top ranked and low ranked by linear regression model and decision trees. Simulation results established the efficacy of linear regression model over decision trees.

B. Machine Learning for Charging Demand Prediction

Predicting the charging load accurately is crucial for charging infrastructure planning and large-scale adoption of EVs. In [72], the authors presented a novel scheme for predicting aggregated load demand of buildings in presence of EVs by utilizing a methodology based on feature selection and enhanced SVM. In [73], the authors predicted the charging load of UCLA campus by applying modified pattern sequence-based technique. In [74], the authors used deep learning approach for estimating multi-scale EV charging demand. Also, in [75], [76] an enhanced deep learning-based approach is used for charging load prediction. In [77], the authors used hybrid ant lion algorithm and deep learning for charging demand prediction. In [78], the authors proposed a hybrid KDE using both Gaussian- and Diffusion-based KDE (GKDE and DKDE) to predict the stay duration and charging demand of EVs. In [79], authors employed a Generalized Regression Neural Network (GRNN) model for prediction of charging load. In [80], authors predicted the charging demand of electric bus charging stations by SVM and wolf pack algorithm. In [81], the authors compared the performance of different deep learning approaches on the charging demand prediction problem and concluded that long-short-term memory (LSTM) method performs the best by reducing over 30% forecasting error. In [82], the authors used a regression model for predicting the charging load. In [83], the authors compared the performance of time series approach with machine learning techniques such as random forest and regression model on the charging demand prediction problem. Simulation results established the efficacy of machine learning techniques over time series approach. In [84], the authors used ensemble learning to predict the household EV charging demand. Ensemble learning is a form of machine learning technique that is developed from the evaluation of results from different machine learning models. In the aforesaid work, the ensemble learning model is based on the results of Random Forest, Gradient Boosting, Adaptive Boosting, Regression. In [85], the authors used k nearest neighbor method for charging demand prediction.

C. Machine Learning for Charging Scheduling

The management of charging activities in the charging stations is important to avoid sudden increase in peak load demand. In [86], the authors considered operational benefit of the EVs by giving stress to Vehicle to Grid (V2G) and scheduled the charging of EVs in charging station by using reinforcement learning. In [87], the authors proposed a demand response method for long term charging cost reduction and provided a charging schedule for EVs. The solution methodology was based on reinforcement learning. In [88], the authors proposed a constrained EV charge scheduling strategy and utilized reinforcement learning for solving it. In [89], the authors formulated the charging scheduling as a NP hardness problem and solve the same by reinforcement learning. In [90], the authors proposed Artificial Neural Network (ANN) for solving charging scheduling and suggested a smart pricing strategy to be adopted in charging stations. In [91], the authors identified the best charging time for EVs in a fast charging station integrated with smart grid by Q learning method. In [92], [93], the authors solved the scheduling of charging

activity by reinforcement learning. In [94], [95], the authors used multi-agent reinforcement learning for charging scheduling and proposed a dynamic pricing strategy. In [96], the authors proposed a reinforcement-learning based approach for optimizing charging scheduling and pricing strategies that maximize the system objective of a public EV charging station. In [97], the authors used reinforcement learning to propose charging scheduling for electric buses in a charging station in a smart grid environment.

D. Machine Learning for Charger Utilization Computation

Estimating the charger utilization rate is essential for future expansion of charging infrastructure. In [98], the authors predicted the usage of EV charging station by using ANN. In [99], the authors used linear regression model for computing the charger idle time for a dataset of Netherland. In [99], the authors used linear regression model to predict charger utilization rate thereby assuming a nonlinear charging profile

V. SUMMARY

A summary of the research works reported in the previous section is presented systematically in Table III. Further, quantitative analysis of the reported literature is presented in Fig.3. From Fig.3, it is clear that machine learning techniques have their highest applications in charging demand prediction problem.

TABLE III. SUMMARY OF RESEARCH WORKS ON USE OF MACHINE LEARNING FOR CHARGING INFRASTRUCTURE PLANNING

Ref	Year	Problem	Technique
[68]	2019	Charging station placement	Hybrid GA RL
[69]	2018	Charging station placement	Hierarchal clustering
[71]	2019	Charging station placement	Linear regression model and decision trees
[72]	2018	Charging Demand prediction	SVM
[73]	2014	Charging Demand prediction	Modified pattern sequence
[74]	2019	Charging Demand prediction	Deep learning
[75]	2017	Charging Demand prediction	Deep learning
[76]	2019	Charging Demand prediction	Deep learning
[77]	2018	Charging Demand prediction	Hybrid ant lion and deep learning
[78]	2018	Charging Demand prediction	Hybrid KDE
[79]	2020	Charging Demand prediction	GRNN
[80]	2018	Charging Demand prediction	SVM
[81]	2019	Charging Demand prediction	Deep learning
[82]	2020	Charging Demand prediction	Regression model
[83]	2019	Charging Demand prediction	Random forest and regression model
[84]	2018	Charging Demand prediction	Ensemble learning
[85]	2014	Charging Demand	KNN

		prediction	
[86]	2019	Charging scheduling	Reinforcement learning
[87]	2020	Charging scheduling	Reinforcement learning
[88]	2019	Charging scheduling	Reinforcement learning
[89]	2019	Charging scheduling	Reinforcement learning
[90]	2020	Charging scheduling	ANN
[91]	2018	Charging scheduling	Reinforcement learning
[92]	2020	Charging scheduling	Reinforcement learning
[93]	2018	Charging scheduling	Reinforcement learning
[94]	2020	Charging scheduling	Reinforcement learning
[95]	2019	Charging scheduling	Reinforcement learning
[96]	2019	Charging scheduling	Reinforcement learning
[97]	2019	Charging scheduling	Reinforcement learning
[98]	2018	Charger utilization	ANN
[99]	2019	Charger utilization	Linear regression model
[100]	2019	Charger utilization	Linear regression model

VI. CONCLUSION

Large scale deployment of EVs call for availability of sustainable charging infrastructure. This work systematically reports the machine learning applications for solving charging infrastructure planning. A qualitative as well as quantitative analysis of the research works of this arena are provided in this work. It is observed that machine learning algorithms have their highest applications in charging demand prediction followed by charging scheduling. It is expected that this work will attract researchers working in the area of e mobility, optimization, machine learning, power and energy. Our future work will address some of the key aspects such as:

- Use of machine learning for computing charging hotspots
- Comparison of the performance of machine learning with heuristics and metaheuristics on charging infrastructure planning problem

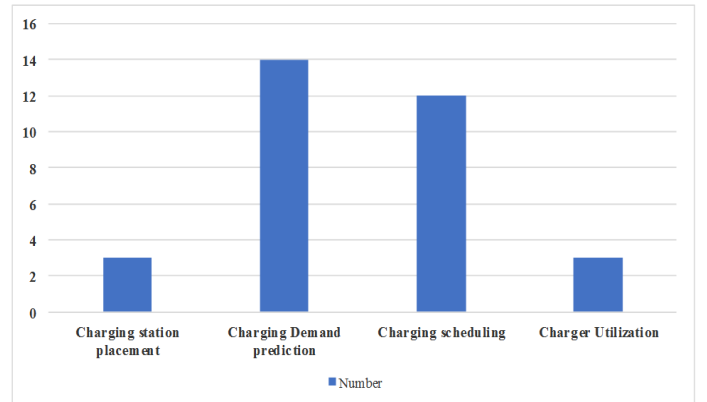


Fig. 3. Quantitative analysis of reported research works

ACKNOWLEDGEMENT

This research is supported by European Research Consortium of Informatics and Mathematics (ERCIM)

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